

# HYEONBEEN LEE

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## EDUCATION

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**Virginia Tech**, Blacksburg, VA, USA

Aug. 2026 — May 2031 (Expected)

Ph.D. in Mechanical Engineering (Advisor: [Simon Stepputtis](#))Research Focus: Vision-Language-Action (VLA), Contact-rich manipulation, Real-time controller**Kyung Hee University**, Seoul, South Korea

Mar. 2022 — Feb. 2024

M.Eng. in Mechanical Engineering (Advisor: [Jin-Gyun Kim](#))

GPA: 3.93/4.0

Thesis: ‘Composite neural network with differential propagation for modeling impulsive nonlinear dynamic systems’**Kyung Hee University**, Seoul, South Korea

Mar. 2015 — Feb. 2022

B.Eng. in Mechanical Engineering (Advisor: [Jin-Gyun Kim](#), [Shinkyu Jeong](#))

(Upper Div.: 3.88/4.0) GPA: 3.52/4.0

Thesis: ‘Data-driven aerodynamic coefficient prediction using deep neural network and PARSEC airfoil parameterization’

(\* Military leave of absence: May 2017 — Feb. 2019)

## PUBLICATIONS

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### Journal Papers

4. **H. Lee**, M. Jung, T.K. Yeu, J.B. Han, D. Park, S. Stepputtis, J.G. Kim (2026), ‘Frequency-aware decomposition learning for sensorless wrench estimation in vibration-rich robotic contact’, submitted to *Robotics and Computer-Integrated Manufacturing*. [[arXiv](#)] [[code](#)] [[dataset](#)]
3. B. Koo, M. Jung, **H. Lee**, T. Yeu, J.G. Kim, J.B. Han, Y. Lee, D. Park (2026), ‘Buoyancy-integrated hybrid reaction force estimation method with real-time haptic feedback for underwater hydraulic manipulation’, submitted to *IEEE Robotics and Automation Letters*.
2. **H. Lee**, S. Han, H.S. Choi, J.G. Kim (2024), ‘cNN-DP: Composite neural network with differential propagation for impulsive nonlinear dynamics’, *Journal of Computational Physics (WoS IF Top 2.5% in Physics, Mathematical)*, 496, 112578. [[paper](#)] [[code](#)]
1. S. Han\*, G.E. Jeong\*, **H. Lee**, W.S. Choi, J.G. Kim (2023), ‘Multi-body dynamics model for spent nuclear fuel transportation system under normal transport test conditions’, *Nuclear Engineering and Technology (WoS IF Top 13.7% in Nuclear Science & Technology)*, 55(11), 4125-4133. (\*Co-first authors) [[paper](#)]

### Conference Proceedings & Presentations

6. J.H. Han, J.B. Han, S.S. Kim, M.H. Kim, Y.H. Kim, **H. Lee**, J.G. Kim, T.K. Yeu. “Digital twin model of underwater construction robot for real-time grinding simulation”, *7th International Conference on Multibody System Dynamics*, Madison, WI, USA, Jun. 2024 [[abstract](#)]
5. **H. Lee**, J. Han, T.K. Yeu, J.G. Kim. “Real-time multi-horizon reaction force forecasting of marine robot using interpretable Transformer”, *Korean Society of Mechanical Engineers* (Oral Presentation), Incheon, South Korea, Nov. 2023 [[abstract](#)]
4. **H. Lee**, S. Han, H.S. Choi, J.G. Kim. “Meta-modeling of nonlinear impulsive dynamics using composite neural network model with differential propagation”, *Korean Society of Mechanical Engineers* (**Excellent Paper Award**, Oral Presentation, \* Extended works from IMAC 2023), Busan, South Korea, May 2023. [[abstract](#)]
3. **H. Lee**, S. Han, H.S. Choi, J.G. Kim. “Composite neural network with differentiation propagation for modeling nonlinear impulsive dynamics”, *Conference on Dynamics and Control, Korean Society of Mechanical Engineers* (Oral Presentation), Jeju, South Korea, Mar 2023 [[abstract](#)]
2. **H. Lee**, S. Han, H.S. Choi, J.G. Kim. “Composite neural network framework for modeling impulsive nonlinear dynamic responses”, *41st International Modal Analysis Conference (IMAC)* (Oral Presentation), Austin, TX, USA, Feb. 2023 [[paper](#)]
1. **H. Lee**, J.G. Kim. “Development of multibody dynamics trailer model using normal transportation test data and DNN based surrogate model generation”, *Korean Society for Noise and Vibration Engineering* (Oral Presentation), Jeju, South Korea, Dec 2022

## AWARDS & HONORS

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- **Excellent Paper Award** Korean Society of Mechanical Engineers, 2023  
– Awarded for a first-author presentation at KSME 2023 conference.
- **Brain Korea 21 Plus Scholarship** Graduate School, Kyung Hee University, 2022 — 2023  
– Received \$16K in graduate funding during M.Eng. studies.
- **Graduate Assistantship** Graduate School, Kyung Hee University, 2022 — 2023  
– Received \$11K through a graduate assistantship during M.Eng. studies.
- **Academic Superiority Scholarship** Dept. of Mechanical Engineering, Kyung Hee University, Spring 2021  
– Ranked 1st in the GPA-based departmental merit review and received a full tuition waiver valued at \$3K.

## RESEARCH EXPERIENCES

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### Virginia Tech, Department of Mechanical Engineering

Thinking Embodied Agents (TEA) Lab., advised by *Dr. Simon Stepputtis*

Graduate Research Assistant

Aug. 2026 — Present

- Developing hierarchical Vision-Language-Action (VLA) controllers and sensorless contact perception methods for real-time robotic manipulation.

### Kyung Hee University, Graduate School

Modeling & Simulation Lab., advised by *Dr. Jin-Gyun Kim*

Research Assistant

Mar. 2024 — Apr. 2026

- Developed a frequency-aware decomposition learning method for robotic wrench forecasting. Submitted a first-authored manuscript to the *IEEE/ASME Transactions on Mechatronics (T-MECH)*.
- Refined the forecasting model for real-time robotic wrench estimation for KRISO.

### KAIST, Kim Jaechul Graduate School of AI

Cognitive Learning for Vision & Robotics Lab., advised by *Dr. Joseph J. Lim*

Research Intern

Sep. 2023 — Nov. 2023

- Conducted an in-depth review of skill-prior regularized reinforcement learning (SPiRL) and delivered a 90-minute presentation to lab members and faculty.

### Kyung Hee University, Graduate School

Modeling & Simulation Lab., advised by *Dr. Jin-Gyun Kim*

Graduate Research Assistant

Mar. 2022 — Feb. 2024

- Proposed a composite neural network architecture for learning complex dynamics from multi-order derivatives. Published a first-author article in the *Journal of Computational Physics (JCOMP)*.
- Developed a multibody dynamics vehicle model for the Korea Atomic Energy Research Institute (KAERI). Published a co-authored article in the *Nuclear Engineering and Technology (NET)*.
- Designed a learning-based model to forecast the contact-rich wrench of a hydraulic manipulator for the Korea Research Institute of Ships and Ocean Engineering (KRISO).

### Kyung Hee University, School of Mechanical Engineering

Modeling & Simulation Lab., advised by *Dr. Jin-Gyun Kim*

Undergraduate Research Assistant

Jan. 2021 — Feb. 2022

Aerodynamic Design Exploration & Big Data Analysis Lab, advised by *Dr. Shinkyu Jeong*

Sep. 2020 — Jun. 2021

- Investigated data-driven methods for flexible multibody dynamics and aerodynamics.
- Led a thesis project on learning-based aerodynamic design optimization.

## SKILLS

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- **Programming Languages:** Python · MATLAB ·  $\text{\LaTeX}$
- **Software & Frameworks:** PyTorch · Git · Docker · ROS · RecurDyn · Gazebo · MuJoCo
- **Scientific Computing:** Flexible multibody dynamics · Finite element methods · Numerical analysis
- **Natural Languages:** Korean (Native) · English (TOEFL iBT 111/120) · Japanese (Advanced proficiency)

## INVITED SEMINARS

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2. “On the data-driven modeling of contact- and vibration-rich dynamics”, invited seminar, IMBD Lab, Chungnam National University, Daejeon, South Korea, Apr. 16, 2026 [\[keynote\]](#)
1. “On the data-driven modeling of contact- and vibration-rich dynamics”, invited seminar, Modeling & Simulation Lab, Kyung Hee University, Yongin, South Korea, Apr. 14, 2026 [\[keynote\]](#)

## TEACHING & MENTORSHIP

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### Research Mentor

— Guiding junior graduate students on robotics and machine learning.

Modeling & Simulation Lab., Kyung Hee University

Mar. 2025 — Jul. 2026

### Teaching Assistant, ME380 Systems Dynamics

— Lectured on multibody dynamics with theoretical derivation and its computational implementation.

Dept. of Mechanical Engineering, Kyung Hee University

Mar. 2022 — Jun. 2023

## COMMUNITY SERVICE

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### Lead Administrative Assistant

48th College of Engineering Student Council

ROK-U.S. Combined Marine Corps Interpreter

Dept. of Mechanical Engineering, Kyung Hee University, 2022 — 2024

Kyung Hee University, Feb 2019 — Jan. 2020

1st Marine Div., ROKMC, Sep 2017 — Feb. 2019

## REFERENCES

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- **Jin-Gyun Kim**, Ph.D.

Associate Professor at Department of Mechanical Engineering, Kyung Hee University ([jingyun.kim@khu.ac.kr](mailto:jingyun.kim@khu.ac.kr))

- **Hee-Sun Choi**, Ph.D.

Assistant Professor at Department of Artificial Intelligence and Robotics, Sejong University ([heesunchoi@sejong.ac.kr](mailto:heesunchoi@sejong.ac.kr))

- **Seongji Han**, Ph.D.

Assistant Professor at Department of Mechatronics Engineering, Chungnam National University ([seongji.han@cnu.ac.kr](mailto:seongji.han@cnu.ac.kr))